4. Computation of Coarse Cohomology

4.0. Reminder

* $f: X \to Y$ is "controlled" if $\forall E \in \mathcal{E}_X$, $(\mathrm{id}_X \times f)(E) \in \mathcal{E}_Y$.

  = "proper" if $\forall B \subseteq Y$ bdd, $f^{-1}(B)$ is bdd in X

  = "coarse" if it is controlled and proper.

Coarse ; Coarse$/_{\simeq}$

$f: X \to Y$ is a c.equivalence if $\exists g: Y \to X \in \mathrm{Coarse}$ s.t.

$gf \simeq \mathrm{id}_X$ $\quad$ $fg \simeq \mathrm{id}_Y$.

4.1. Coarse Cohomology

The defn given here is equivalent to the original by John Roe : "Lectures on Coarse Geometry", §5.

* $X \in \mathrm{Coarse}$, G abelian group.

$C^*(X; G)$ "baric cochain complex" (of X w/ coeff. in G)

$C^n(X; G) = \{ \varphi : X^{n+1} \to G \}$ $\quad$ $\downarrow \mathcal{D}_\bullet X$ $\quad$ $\mathcal{D}_n X = X^{n+1}$

$(x_0, \dots, x_n) \mapsto \varphi(x_0, \dots, x_n) \in G$

n-simplices ($n+1$ vertices)

$\delta : C^n(X; G) \to C^{n+1}(X; G)$;

$\varphi : X^{n+1} \to G \mapsto \delta\varphi : X^{n+2} \to G$;

$(x_0, \dots, x_{n+1}) \mapsto \sum_{i=0}^{n+1} (-1)^i \varphi(x_0, \dots, \hat{x_i}, \dots, x_{n+1})$

σ $\quad$ i-th face of σ

$\delta^2 = 0$;

$C^*(X)$ is contractible up to degree zero ;

* Defn : A cochain $\varphi : X^{n+1} \to G$ is a "coarse cochain" if

$\forall E \in \mathcal{E}_X$, $E[\Delta] \cap \mathrm{supp}\,\varphi$ is bdd

$\Delta = \{(x, \dots, x) \mid x \in X\} \subseteq X^{n+1}$;

$E[\Delta] = \{\sigma = (x_0, \dots, x_n) \mid \exists x \in X \text{ s.t. } (x, x_i) \in E \ \forall i = 0, \dots, n\}$

$E[\Delta] \cap \mathrm{supp}\,\varphi \subseteq X^{n+1}$

"$E[\Delta]\cap \mathrm{supp}\,\varphi$ bded" means $\exists \tilde{x}\in X$ and $G\in\mathcal{E}_X$ s.t.

$$E[\Delta]\cap \mathrm{supp}\,\varphi \subseteq G[(\tilde{x},\dots,\tilde{x})] = \{\sigma=(x_0,\dots,x_n) \mid (\tilde{x},x_i)\in G \ \forall i=0,\dots,n\}$$

Let

$$CX^n(X;G) = \{\varphi : X^{n+1}\to G \mid E[\Delta]\cap\mathrm{supp}\,\varphi \text{ is bded } \forall E\in\mathcal{E}_X\}$$

$$CX^n(X)\le C^n(X). \qquad \delta(CX^n(X))\subseteq CX^{n+1}(X)$$

- Lemma: $CX^*(X)$ is a subcomplex of $C^*(X)$.

Let $\varphi\in CX^n(X)$. Let $E\in\mathcal{E}_X$. We want to check that $E[\Delta]\cap\mathrm{supp}(\delta\varphi)$ is bded. We know $E[\Delta]\cap\mathrm{supp}(\varphi)$ is bded; $\exists\,\tilde{x}\in X$ and $G\in\mathcal{E}_X$ s.t. $E[\Delta]\cap\mathrm{supp}\,\varphi\subseteq G[(\tilde{x},\dots,\tilde{x})]$.

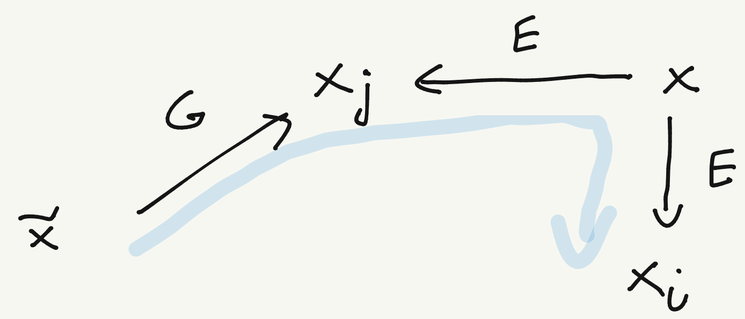
Let $\sigma\in E[\Delta]\cap\mathrm{supp}(\delta\varphi)$; $\sigma=(x_0,\dots,x_{n+1})$: Let $x\in X$ s.t. $(x,x_j)\in E \ \forall j=0,\dots,n+1$

Since $(\delta\varphi)(\sigma)\neq 0$, $\exists i=0,\dots,n+1$ s.t. $\varphi(\sigma_i)\neq 0$;

$\sigma_i=(x_0,\dots,\hat{x_i},\dots,x_{n+1})$. Since $\sigma\in E[\Delta]$, $\sigma_i\in E[\Delta]$;

$\sigma_i\in E[\Delta]\cap\mathrm{supp}\,\varphi\subseteq G[(\tilde{x},\dots,\tilde{x})]$;

$\forall j=0,\dots,\hat{i},\dots,n+1$, $(\tilde{x},x_j)\in G$. Let $j\neq i$,



$(\tilde{x},x_i)\in E\circ E^{-1}\circ G$

$(\tilde{x},x_j)\in G\cup(E\circ E^{-1}\circ G) \quad \forall j=0,\dots,\hat{i},\dots,n+1$.

$\in E[\Delta]\cap\mathrm{supp}(\delta\varphi)$ is bded.

L___.

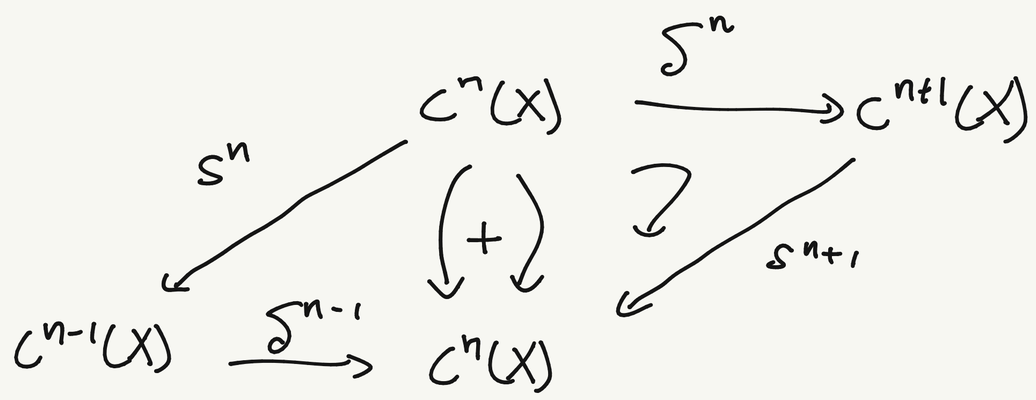
□

Contractibility of $C^*(X)$

Assume $X \neq \emptyset$. Choose a basepoint $p \in X$.

$$s^n : C^n(X) \longrightarrow C^{n-1}(X) \quad ; \quad n \geq 1$$

$$\varphi : X^{n+1} \to G \mapsto (s\varphi) ; (x_0, \dots, x_{n-1}) \mapsto \varphi(p, x_0, \dots, x_{n-1})$$



$$s^{n+1}\delta^n + \delta^{n-1}s^n = id_{C^n(X)} \quad \text{for } n \geq 1.$$

$$0 \to G \xrightarrow{\varepsilon} C^0(X) \longrightarrow C^1(X) \longrightarrow C^2(X) \longrightarrow \cdots$$

$\widetilde{C}^*(X)$ ↓ $\widetilde{H}^*(X)$

- **Defn:** $HX^*(X)$ "coarse cohomology of X w/ coeff. in G^n" is defined as the cohomology of $\mathcal{C}^*(X; G)$.

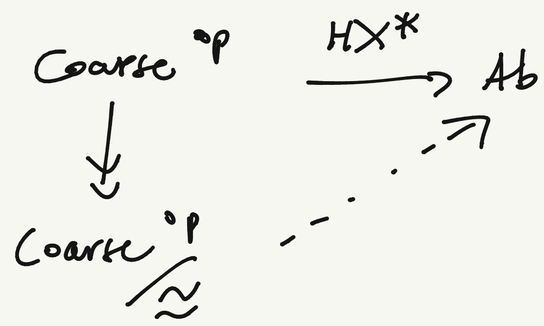
- **Prop:**

$$HX^* : \text{Coarse}^{op} \longrightarrow Ab$$
$$f : X \to Y \rightsquigarrow f^* : HX^*(Y) \longrightarrow HX^*(X)$$

Moreover, HX^* is a "coarse invariant":

$$[f, g : X \rightrightarrows Y \; ; \; f \simeq g] \Rightarrow f^* = g^*$$

Therefore, if f is a coarse equivalence, f^* is an isomorphism.



- $X \underset{g}{\overset{f}{\rightrightarrows}} Y$ in Coarse.

$$f \approx g \overset{\text{def}}{\iff} (f \times g)(\Delta) \in \mathcal{E}_Y$$

"f and g are close"

$$\iff \exists F \in \mathcal{E}_Y \text{ s.t. } (f(x), g(x)) \in F \quad \forall x \in X.$$

$$d_\infty(f,g) < \infty.$$

- How to compute?

4.2: Topological bornological spaces.

- Defn: A bornology on a set X is a family $\mathcal{B}$ of subsets of X s.t.:

i) $X = \bigcup_{B \in \mathcal{B}} B$ $\quad x \in \{x\}$

ii) $B_1, B_2 \in \mathcal{B} \implies B_1 \cup B_2 \in \mathcal{B}$ $\quad \leftarrow$

iii) $B_1 \subseteq B_2 \in \mathcal{B} \implies B_1 \in \mathcal{B}$ $\quad \leftarrow$

Elements of $\mathcal{B}$ are called "bounded sets".

- Example: (X,d); $\mathcal{B} = \{\text{bded sets}\}$ is a bornology.
- Example: (X,τ) top. space. Assume it is loc. cpt. Hausdorff. Then $\mathcal{B} = \{\text{rel. cpt. sets}\}$ is a bornology.
- Defn: Let (X,τ) be a top. space. A bornology $\mathcal{B}$ on X is "compatible w/ τ" if:

i) $B \in \mathcal{B} \implies \overline{B} \in \mathcal{B}$.

ii) $B \in \mathcal{B} \implies \exists N \in \mathcal{B} \cap \tau$ s.t. $B \subseteq N$.

$(X, \tau, \mathcal{B})$ is a topological bornological space.

• Rmk: If $(X, \tau, \mathcal{B})$ is top. born. space, then every rel. cpt set is bounded:

If B is rel. cpt, $\overline{B}$ is compact.

$\forall x \in \overline{B}$, $\exists N_x$ open bded st. $x \in N_x$: $\{N_x \mid x \in \overline{B}\}$ open cover $\overline{B}$. Let $\{N_{x_1}, \dots, N_{x_K}\}$ finite subcover.

$$\overline{B} \subseteq \bigcup_{i=1}^{K} N_{x_i}$$

$$\mathrm{Top}^{\mathrm{lc.Haus}} \hookrightarrow \mathrm{Top\,Born}$$ proper homotopy theory.

• TopBorn is the category where:

Obj: $(X, \tau, \mathcal{B})$ top. born. spaces.

Morphisms: $f: X \to Y$ is a morphism if it continuous and proper

$\forall B \in \mathcal{B}_Y$, $f^{-1}(B) \in \mathcal{B}_X$.

• Bornological cohomology.

$$CB^*(X) \hookrightarrow C^*(X)$$

$CB^n(X) = \{\varphi : X^{n+1} \to G \mid \Delta \cap \overline{\mathrm{supp}\,\varphi}$ is bded$\}$

bornological n-cochains

$CB^*(X)$ is a cochain subcomplex of $C^*(X)$.

The bornological cohomology of X is the cohomology of $CB^*(X)$;

$$HB^*(X) = \frac{\ker\left(\delta \mid_{CB^*(X)}\right)}{\mathrm{Im}\left(\delta \mid_{CB^{*-1}(X)}\right)}$$

- Prop: HB^* : $\mathrm{TopBorn}^{op} \longrightarrow Ab$

  $f : X \to Y \rightsquigarrow HB^*(f) : HB^*(Y) \longrightarrow HB^*(X)$.

HX^* related to HB^* ?

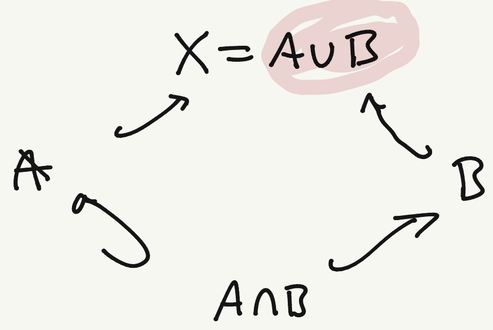
How ?

$HX^* : \mathrm{Coarse}^{op} \to Ab$

Mayer-Vietoris seq. in the literature

$HB^* : \mathrm{TopBorn}^{op} \to Ab$

Mayer-Vietoris ?



$X = \mathring{A} \cup \mathring{B}$

$A \cup B = X \rightsquigarrow$ (relax) $A \cup B$ is "coarsely dense" on X

$\exists E \in \mathcal{E}_X$ s.t. $E[A \cup B] = X$

$\{ y \in X \mid \exists x \in A \cup B$ for which $(x,y) \in E \}$

•) "Coarse Cohomology and Index Theory of Complete Riem. mnflds" Chapters 2, 3

•) "Lectures on Coarse Geometry" Chapter 5.

John Roe

$$HX^*(M) \longrightarrow H_c^*(M) \longrightarrow H^*(M)$$